

AI INTERVIEW & SKILL PASSPORT ENGINE

Project Overview

AI-powered interview preparation platform evaluating candidate resumes, conducting technical viva assessments, and generating verifiable Skill Passports.

Problem Statement:	Candidates lack realistic technical interview practice with instant feedback, while recruiters struggle to verify real skill levels.
Solution Implemented:	Architected an interactive evaluation system combining OpenAI LLM prompts, client-side Python execution via Pyodide, and automated scoring.
My Technical Role:	Creator & AI Engineer

Main Features & Technical Stack

Core Features:	Technologies Used:
- Resume parsing & natural language skill extraction - Dynamic technical viva & domain interview question generation - Browser-based Pyodide Python execution sandbox - Evidence-backed shareable Skill Passport generation	React, TypeScript, FastAPI, Python, PostgreSQL, Pyodide WASM, OpenAI API, Vercel
	Architecture Flow:
	React Client → FastAPI Gateway → Pyodide WASM Engine → OpenAI API → PostgreSQL DB → Vercel

Technical Challenge & Results

Key Engineering Challenge: Eliminated LLM grading inconsistencies via strict JSON evaluation rubrics and ran code safely on client via WebAssembly.

Validated Results: Sub-3-second technical answer evaluation with automated verifiable candidate Skill Passports.